Tsung-Ju Lee (李宗儒)

Assistant Professor at the Department of Mathematics, National Cheng Kung University

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Research Interests

- Algebraic geometry and complex differential geometry, especially focusing on their applications to Calabi–Yau varieties and their moduli problems.

Education

Aug 2010 – June 2016 **National Taiwan University**, Ph.D. in Mathematics – Taipei, Taiwan

- Advisors: Professors Chin-Lung Wang and Hui-Wen Lin
- Dissertation: *Geometry arising from periods of Calabi–Yau degenerations and transitions*

Aug 2005 – Jan 2009 **National Taiwan University**, B.S. in Mathematics – Taipei, Taiwan

Experience

Sept 2023 – present **Assistant Professor**, National Cheng Kung University – Tainan, Taiwan

Sept 2018 – Aug 2023 **Postdoctoral Fellow**, Center of Mathematical Sciences and Applications, Harvard University – Cambridge, MA

Feb 2018 – Aug 2018 **Postdoctoral Fellow**, National Taiwan University – Taipei, Taiwan

Nov 2016 – Oct 2017 **Airman**, Ministry of Defense, Taiwan – Taipei, Taiwan

Completed mandatory military service in Taiwan.

Preprints

Oct 2026 **On the hyperplane conjecture**
Tsung-Ju Lee
arxiv.org/pdf/2610.26186 (arXiv)

Oct 2025 **Mirror symmetry for singular double cover Calabi–Yau varieties: quantum test**
Tsung-Ju Lee, Bong H. Lian, Shing-Tung Yau
arxiv.org/abs/2510.05470 (arXiv)

July 2025 **Non-commutative resolutions and pre-quotients of Calabi–Yau double covers**
Tsung-Ju Lee, Bong H. Lian, Mauricio Romo, Leonardo Santilli
arxiv.org/abs/2507.00633 (arXiv)

Jan 2025 **A supplement to the anticanonical volumes of weak Q-Fano threefolds of Picard rank two**
Ching-Jui Lai, Tsung-Ju Lee
arxiv.org/abs/2502.19419 (arXiv)

July 2023 **Non-commutative resolutions as mirror of singular Calabi–Yau varieties**
Tsung-Ju Lee, Bong H. Lian, Mauricio Romo
arxiv.org/abs/2307.02038 (arXiv)

Publications

Mar 2025 **Finite distance problem on the moduli of non-Kähler Calabi–Yau $ \partial\bar{\partial}$-threefolds**
Tsung-Ju Lee
10.1007/s00208-025-03115-8 (Math. Ann. **392** (2025), 1541-1583)

Mar 2025 **Twisted GKZ hypergeometric functions and relative homology**
Tsung-Ju Lee, Dingxin Zhang
10.1007/s00208-024-03058-6 (Math. Ann. **391** (2025), 5553-5580)

Aug 2024 **On a conjecture of Huang–Lian–Yau–Yu**
Tsung-Ju Lee, Bong H. Lian, Dingxin Zhang
10.4310/CAG.2023.v31.n6.a1 (Comm. Anal. Geom. **31** (2023), no. 6, 1341-1352)

May 2024 **Mirror symmetry for double cover Calabi–Yau varieties**
Shinobu Hosono, Tsung-Ju Lee, Bong H. Lian, Shing-Tung Yau
10.4310/jdg/1717356161 (J. Differential Geom. **128** (2024), no. 1, 409-431)

Jan 2024 **SYZ mirror symmetry for del Pezzo surfaces and affine structures**
Siu-Cheong Lau, Tsung-Ju Lee, Yu-Shen Lin
10.1016/j.aim.2024.109488 (Adv. Math. **439** (2024), 109488)

July 2023 **A note on periods of Calabi–Yau fractional complete intersections**
Tsung-Ju Lee
10.1007/s00209-023-03321-70 (Math. Z. **304** (2023), article number 60)

June 2023 **On the complex affine structures of SYZ fibration of del Pezzo surfaces**
Siu-Cheong Lau, Tsung-Ju Lee, Yu-Shen Lin
10.4310/ATMP.2022.v26.n6.a7 (Adv. Theor. Math. Phys. **26** (2023), no. 4, 921-955)

May 2023 **Period domains for gravitational instantons**
Tsung-Ju Lee, Yu-Shen Lin
10.1090/tran/8856 (Trans. Amer. Math. Soc. **376** (2023), 5461-5501)

Feb 2022 **On Calabi–Yau fractional complete intersections**
Tsung-Ju Lee, Bong H. Lian, Shing-Tung Yau
10.4310/PAMQ.2022.v18.n1.a10 (Pure Appl. Math. Q. **18** (2022), no. 1, 317-342)

Nov 2020 **A-hypergeometric systems and relative cohomology**
Tsung-Ju Lee, Dingxin Zhang
10.1142/S0129167X2050113X (Internat. J. Math. **31** (2020), no. 13, 2050113)

July 2019 **Tautological systems under the conifold transition on G(2,4)**
Tsung-Ju Lee, Hui-Wen Lin
10.4310/AJM.2019.v23.n3.a7 (Asian J. Math. **23** (2019), no. 3, 501-526)

July 2018 **A Hodge theoretic criterion for finite Weil–Petersson degenerations over a higher dimensional base**
Tsung-Ju Lee
10.4310/MRL.2018.v25.n2.a14 (Math. Res. Lett. **25** (2018), no. 2, 617-647)

Grants

Aug 2026 – July 2029 **National Science and Technology Council (NSTC) Research Grant**
Grant Number: NSTC 115-2115-M-006-013-MY3

Nov 2023 – July 2026 **National Science and Technology Council (NSTC) Research Grant**
Grant Number: NSTC 112-2115-M-006-016-MY3

July 2020 – June 2023 **AMS Simons Travel Grant**
The AMS Simons Travel Grant is a grant awarded to early-career faculty who have the potential to serve as academic role models in research and education.

Conferences/workshops organized

July 2024 – July 2024 **Geometry crossing singularities – TIMS NTU**

June 2023 – June 2023 **2023 TIMS Seminar in Algebraic Geometry: Calabi–Yau with small parameters – TIMS NTU**

Nov 2022 – Dec 2022 **Representation Theory, Calabi–Yau Manifolds, and Mirror Symmetry – Harvard CMSA**

Invited talks

Apr 2026 **Special Geometry, D-modules, and Modular Forms – NTU TIMS**
Title: *Mirror duality between Calabi–Yau double covers*

Jan 2026 **2025 International Congress of Chinese Mathematicians – Shanghai**
Title: *Mirror symmetry for singular double cover Calabi–Yau varieties: quantum test*

Nov 2025 **Korea-Taiwan Enumerative Geometry Meeting 2025 – NCKU**
Title: *GKZ systems and their solutions*

Dec 2024 **The 1st Cross-Strait Symplectic Geometry Conference – USTC**
Title: *Mirror duality between singular Calabi–Yau varieties: the trivial nef-partition case*

Jan 2024 **Mirror Symmetry and Related Topics – KIAS**
Title: *On periods of Calabi-Yau fractional complete intersections*

Oct 2023 **East Asian Symplectic Conference 2023 – Jeju**
Title: $\partial\bar{\partial}$-lemma for certain non-Kähler CY threefolds

Aug 2022 **Calabi–Yau manifolds and mirror symmetry - Past, Present, and Future - – Gakushuin U**

Title: *On the solutions to GKZ D-modules with fractional parameters*

Nov 2021 **Simons Collaboration on HMS Annual Meeting 2021 – Simons Foundation**

Title: *Mirror duality between Calabi–Yau fractional complete intersections*

Dec 2020 **2020 ICCM Annual meeting – USTC**

Title: *Mirror duality and fractional complete intersection Calabi–Yau spaces*

Languages

Mandarin Chinese

Native speaker

English

Fluent